

DISCRETE-TIME SIGNALS & SYSTEMS

ENSC 429 – Digital Signal Processing

School of Engineering Science
Simon Fraser University
Burnaby, BC, Canada



DISCRETE-TIME SIGNALS

- Discrete-time signals are sequences of numbers:

..., 3.12, 7, -5.2, 2.831, ...

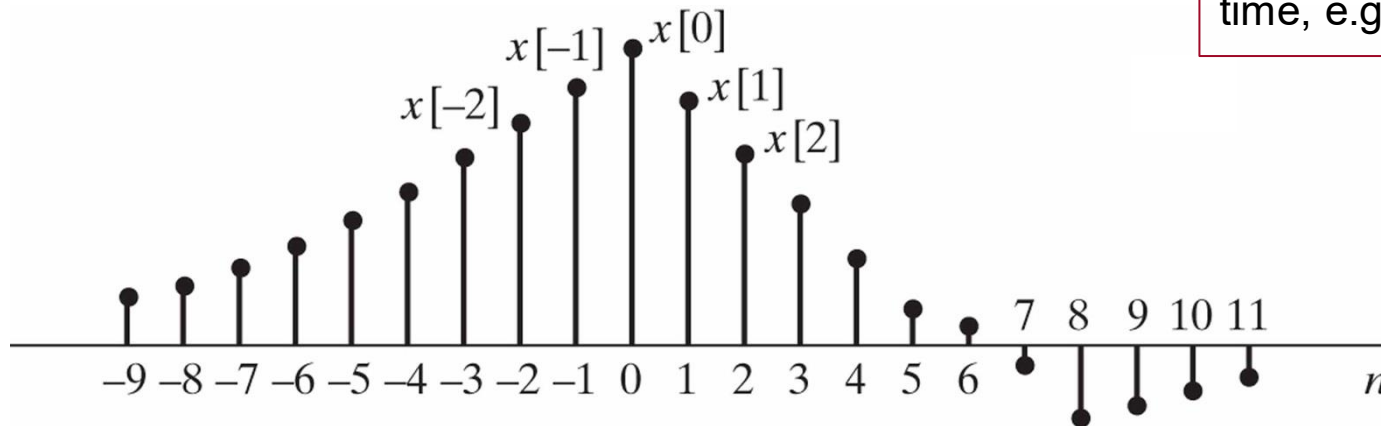
- Notation:

$x[n]$

Index n is called
“discrete time”

- Example:

Square brackets $[\]$ indicate discrete time; round brackets would indicate continuous time, e.g., $x(t)$



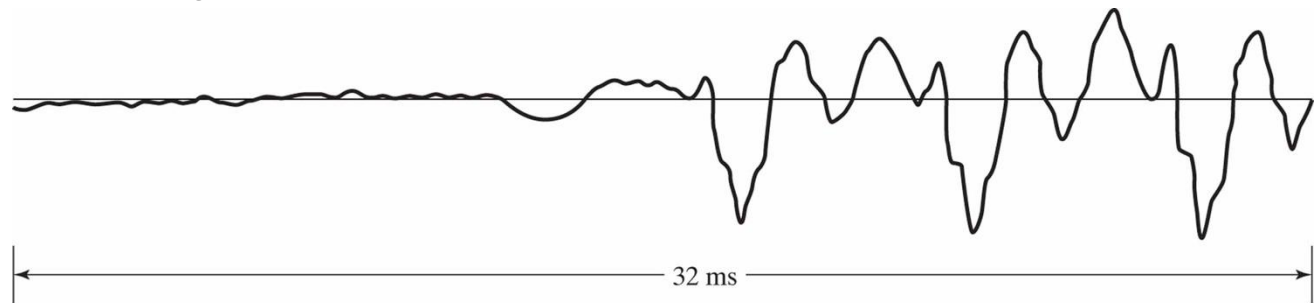
DISCRETE-TIME SIGNALS

- Often, discrete-time signals are obtained by sampling continuous-time signals:

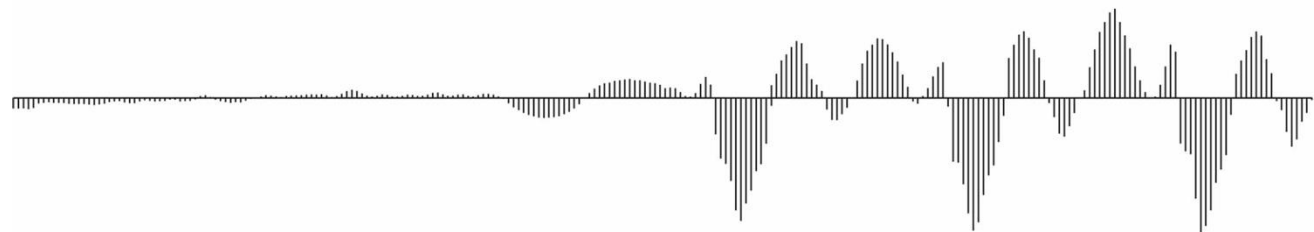
$$x[n] = x(nT)$$

where T is the sampling interval

- Example:

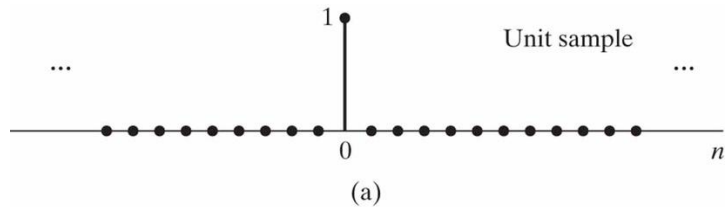


(a)



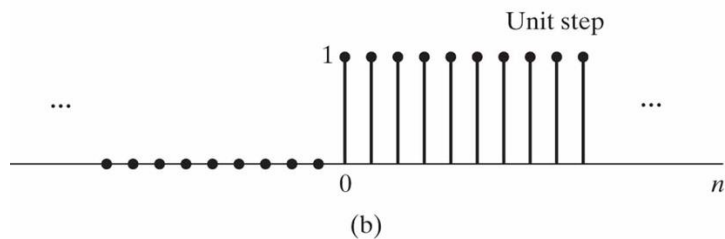
(b)

IMPORTANT DT SIGNALS (SEQUENCES)



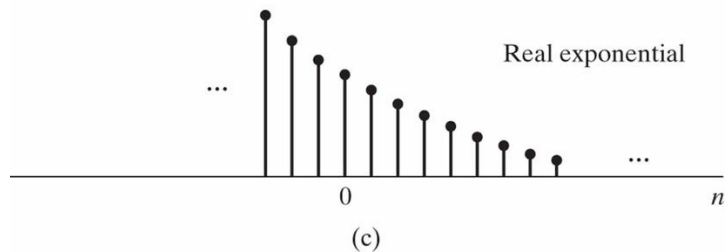
Unit sample (impulse):

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



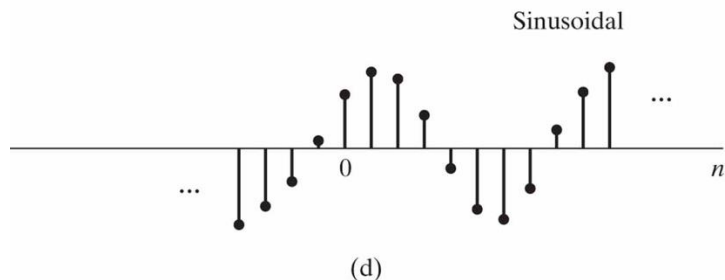
Unit step:

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} = \sum_{k=-\infty}^n \delta[k]$$



Real exponential (A and α real):

$$x[n] = A\alpha^n$$



Sinusoid:

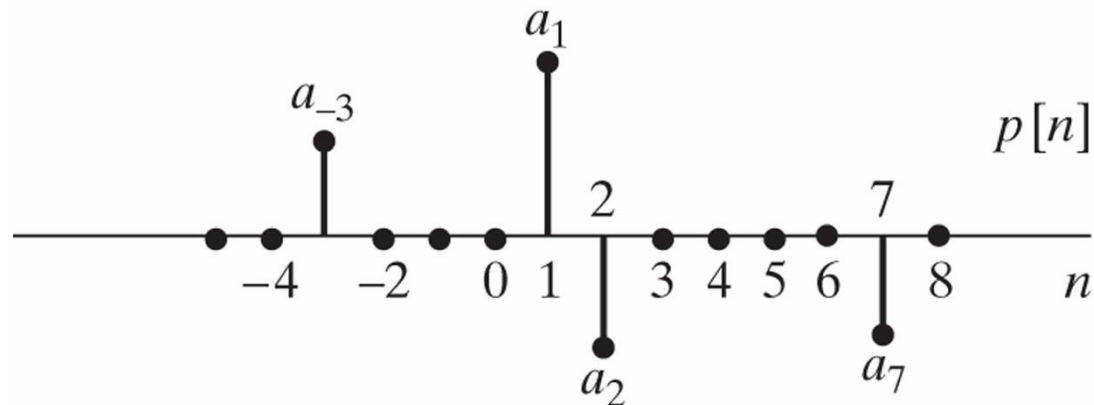
$$x[n] = \cos(\omega_0 n + \phi)$$

DISCRETE-TIME SIGNALS

- Any sequence (discrete-time signal) can be represented as a sum of scaled and delayed impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k]$$

- For example:



$$p[n] = a_{-3}\delta[n + 3] + a_1\delta[n - 1] + a_2\delta[n - 2] + a_7\delta[n - 7]$$

DISCRETE-TIME SIGNALS

- Consider $x[n] = A\alpha^n$ with A and α complex numbers: $A = |A|e^{j\phi}$, $\alpha = |\alpha|e^{j\omega_0}$
- Then we have

$$\begin{aligned}x[n] &= A\alpha^n \\&= |A|e^{j\phi}|\alpha|^n e^{j\omega_0 n} \\&= |A||\alpha|^n e^{j(\omega_0 n + \phi)} \\&= |A||\alpha|^n \cos(\omega_0 n + \phi) + j \cdot |A||\alpha|^n \sin(\omega_0 n + \phi)\end{aligned}$$

- So, a complex exponential is a combination of two sinusoids, scaled by $|A||\alpha|^n$
- ω_0 is the **frequency** (in radians per sample)
- ϕ is the **phase** (in radians)

DISCRETE-TIME SIGNALS

- Consider a discrete-time sinusoid: $x[n] = \cos(\omega_0 n + \phi)$
- If the frequency is $\omega_0 = \omega_1 + 2\pi r$, where r is an integer, then

$$\begin{aligned}x[n] &= \cos(\omega_0 n + \phi) \\&= \cos((\omega_1 + 2\pi r)n + \phi) \\&= \cos(\omega_1 n + 2\pi r n + \phi) \\&= \cos(\omega_1 n + \phi)\end{aligned}$$

Whole multiple
of 2π

- So, although frequencies ω_0 and ω_1 are numerically different, they give rise to exactly the same signal $x[n]$
 - These frequencies are indistinguishable in discrete time
 - Discrete-time frequency is different from continuous-time frequency

DISCRETE-TIME SIGNALS

- Are discrete-time (DT) sinusoids periodic?
- A DT signal $x[n]$ is periodic with period N if and only if

$$x[n] = x[n + N], \quad \text{for all } n$$

- Note that N must be an integer

$$\begin{aligned} x[n] &= \cos(\omega_0 n + \phi) \\ &= \cos(\omega_0(n + N) + \phi) \\ &= \cos(\omega_0 n + \omega_0 N + \phi) \end{aligned}$$

Must be a whole multiple of 2π

- For this to hold, we must have

$$\omega_0 N = 2\pi k, \quad \text{where } k \text{ is an integer}$$

- So, only DT sinusoids whose frequency is of the form below are periodic!

$$\omega_0 = 2\pi \frac{k}{N}, \quad (\text{i.e., } 2\pi \text{ times a rational number})$$

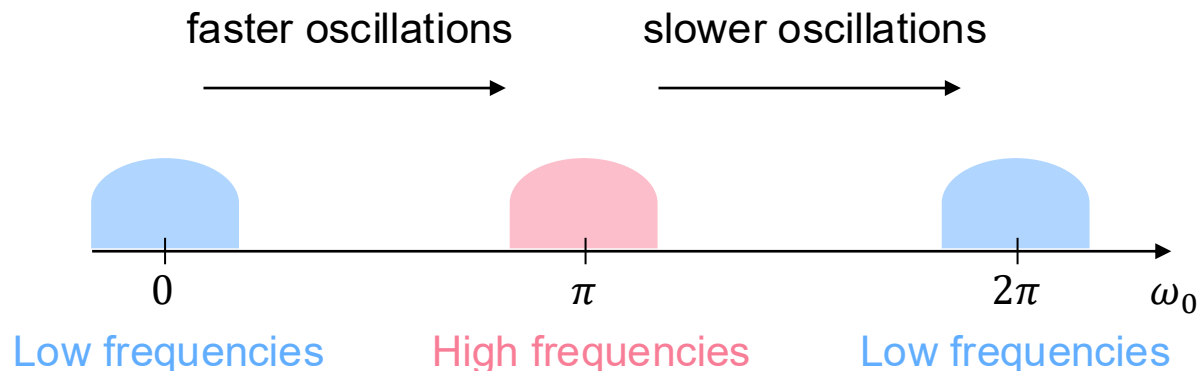
DISCRETE-TIME SIGNALS

There are a few other differences relative to continuous-time sinusoids

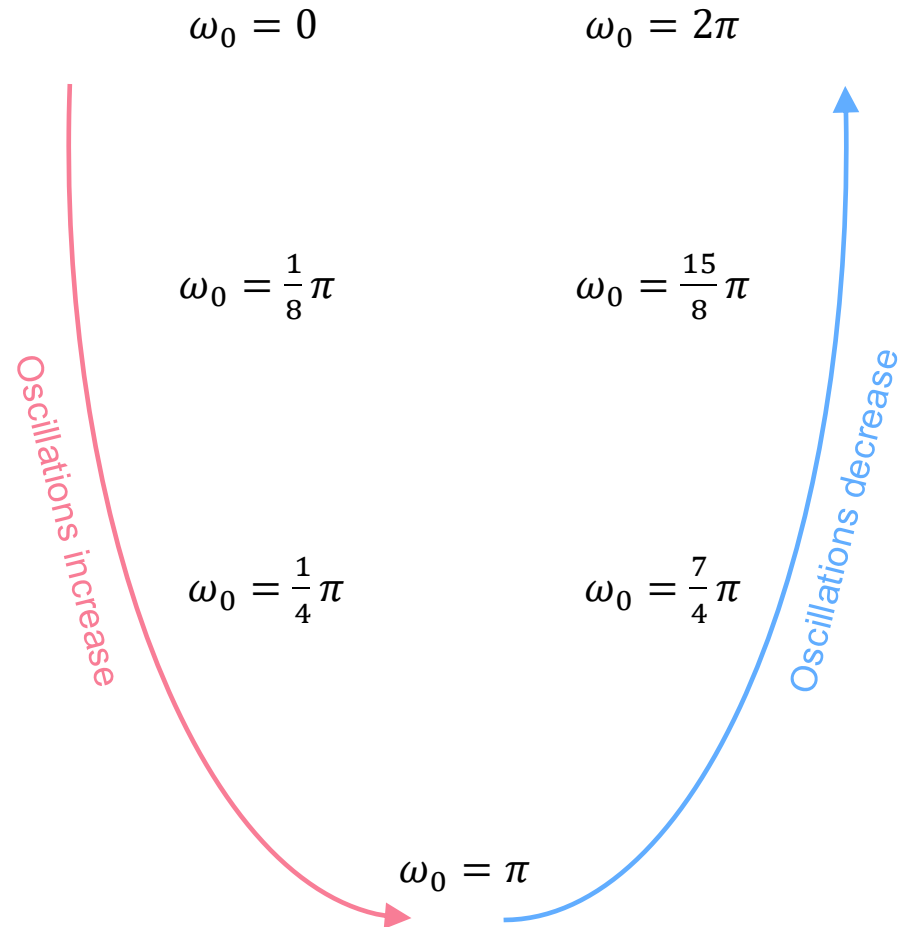
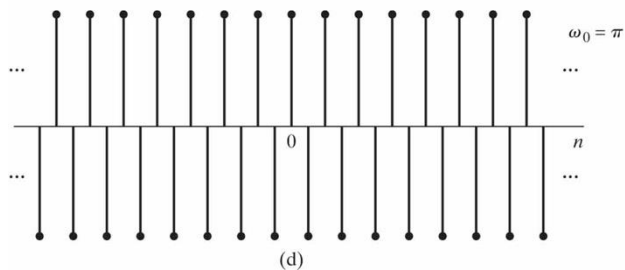
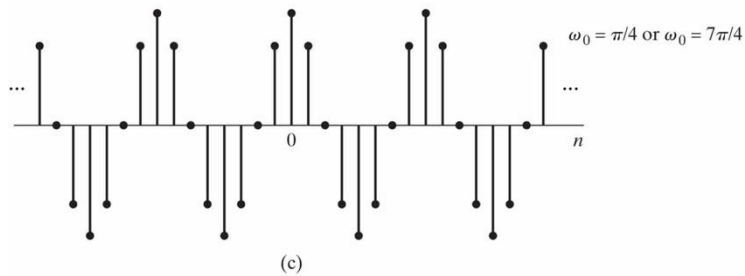
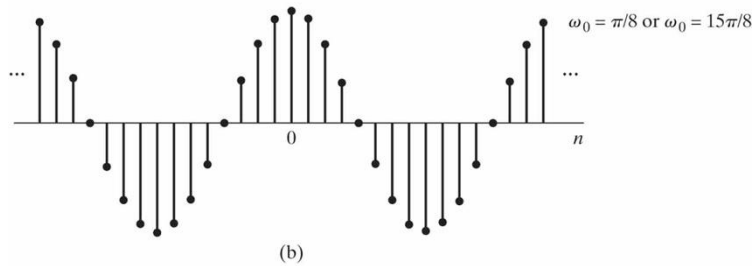
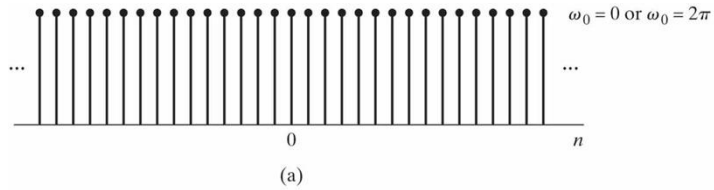
- Given a period N , there are only N distinguishable DT frequencies giving rise to periodic signals with that period:

$$\omega_k = 2\pi \frac{k}{N}, \quad k = 0, 1, \dots, N-1$$

- In continuous-time, when frequency increases, the sinusoidal oscillations become faster. But this is not necessarily the case in DT

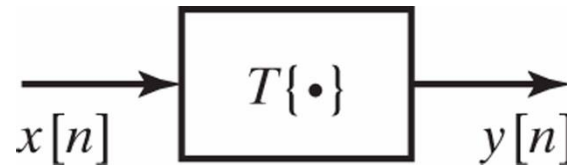


EXAMPLES OF $\cos(\omega_0 n)$



DISCRETE-TIME SYSTEMS

A DT system maps a DT signal to another DT signal



DT system is...	Condition
Memoryless	If, for any n , $y[n]$ depends only on $x[n]$ at the same time n
Causal	If, for any n_0 , $y[n_0]$ depends only on $x[n]$ with $n \leq n_0$
BIBO stable	If, for any bounded input, the output is also bounded. That is, if $ x[n] \leq B_x < \infty$ for all n , then the corresponding output satisfies $ y[n] \leq B_y < \infty$ for all n

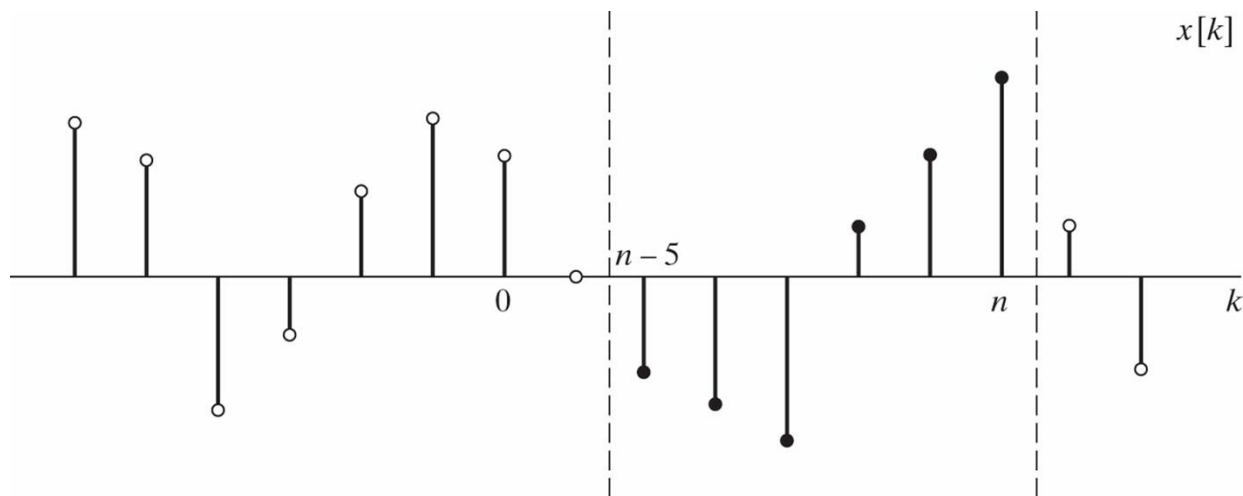
DISCRETE-TIME SYSTEMS

DT system is...	Condition
Linear	If, for any two input signals $x_1[n]$ and $x_2[n]$ and any two constants a_1 and a_2, we have $T\{a_1x_1[n] + a_2x_2[n]\} = a_1T\{x_1[n]\} + a_2T\{x_2[n]\}$ That is, the system satisfies the superposition principle
Time-invariant (or shift-invariant)	If a delay in the system's input causes the same delay in the system's output. That is, if $y[n]$ is the response to $x[n]$, then the response to $x[n - n_0]$ must be $y[n - n_0]$ for any n_0. This must be true for any input $x[n]$ and its corresponding output $y[n]$
Linear-time invariant (LTI) or linear shift-invariant (LSI)	If it is both linear and time-invariant

DISCRETE-TIME SYSTEMS

- Example: moving average

$$y[n] = \frac{1}{6} \sum_{k=0}^5 x[n-k]$$



- Is it memoryless? Causal? BIBO stable? Linear? Time-invariant? [See notes](#)

LINEAR TIME-INVARIANT SYSTEMS

- Any DT signal $x[n]$ can be represented as a sequence of impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

- Let T be a **linear** system and let $h_k[n] = T\{\delta[n - k]\}$ be the response to an impulse applied at time $n = k$
- Then

$$y[n] = T\{x[n]\}$$

$$= T\left\{\sum_{k=-\infty}^{\infty} x[k] \delta[n - k]\right\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] T\{\delta[n - k]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] h_k[n]$$

by the principle
of superposition

LINEAR TIME-INVARIANT SYSTEMS

- If T is also **time-invariant**, and if $h[n] = T\{\delta[n]\}$ is its **impulse response**, then

$$h_k[n] = T\{\delta[n - k]\} = h[n - k]$$

- Using this in the previous equation, we conclude that for **linear time-invariant (LTI) systems**, the input-output relationship is

$$y[n] = T\{x[n]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] h_k[n]$$

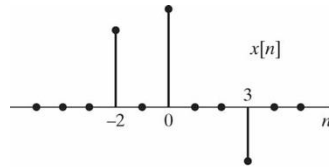
$$= \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

convolution sum

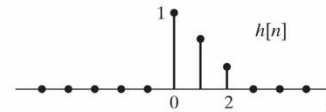
$$= x[n] * h[n]$$

LINEAR TIME-INVARIANT SYSTEMS

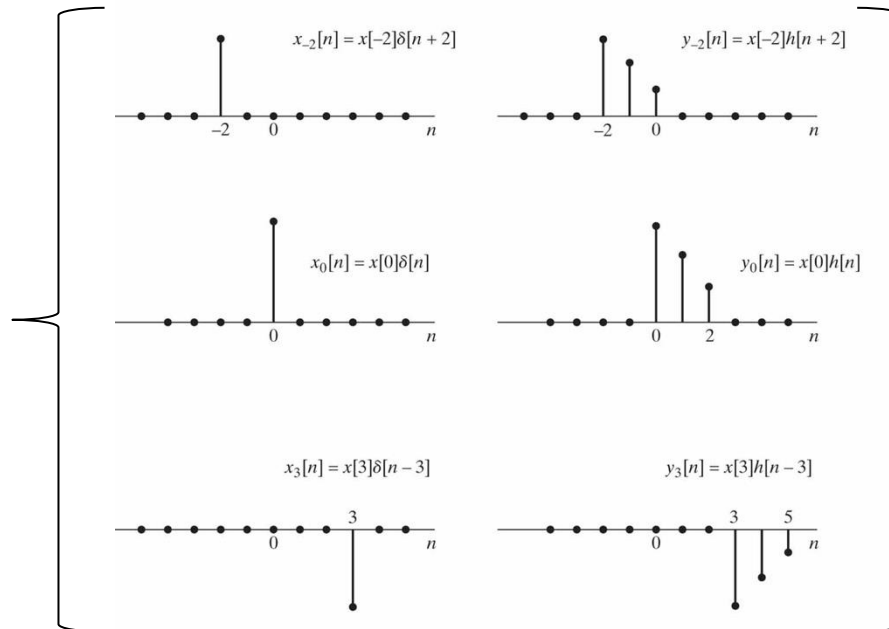
Input signal



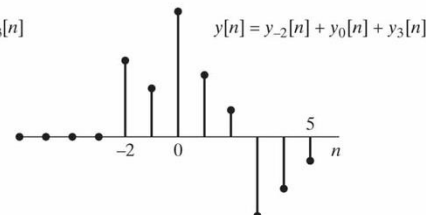
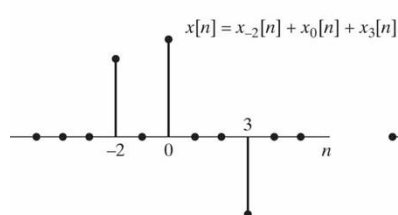
Impulse response



Decomposed into
impulses

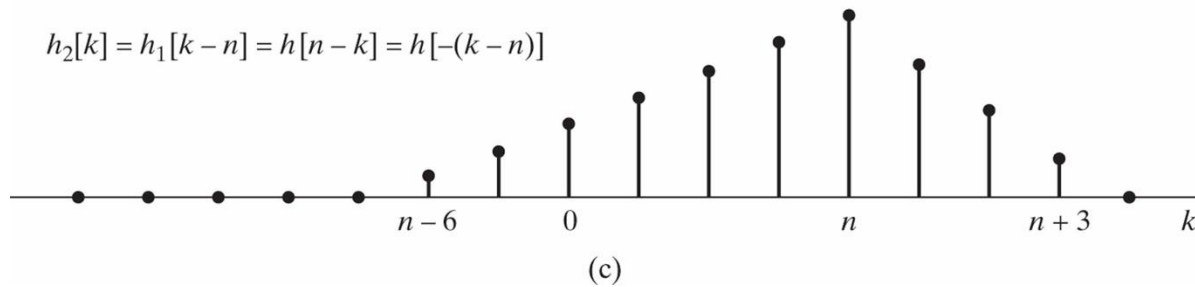
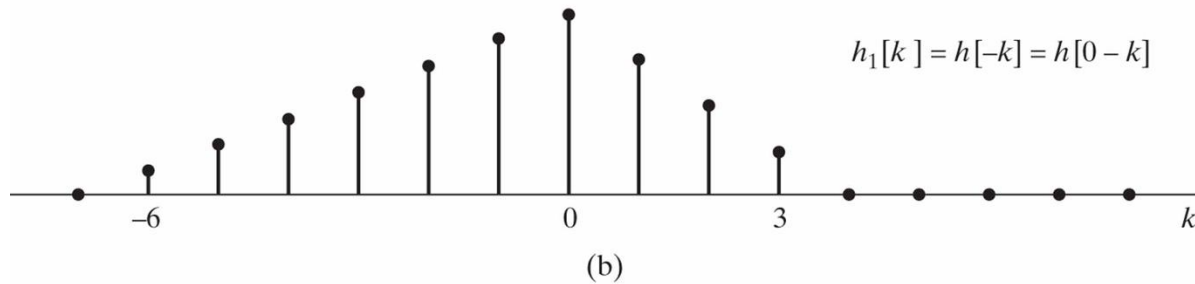
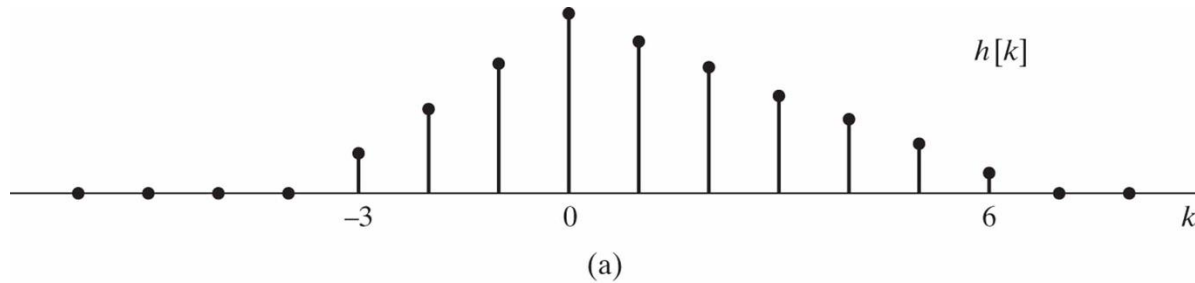


Input signal =
combination of
impulses



Output signal =
combination of
impulse responses

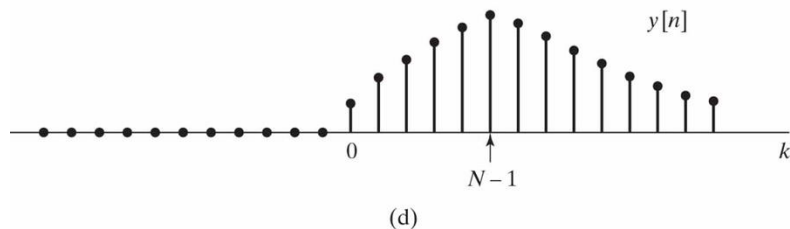
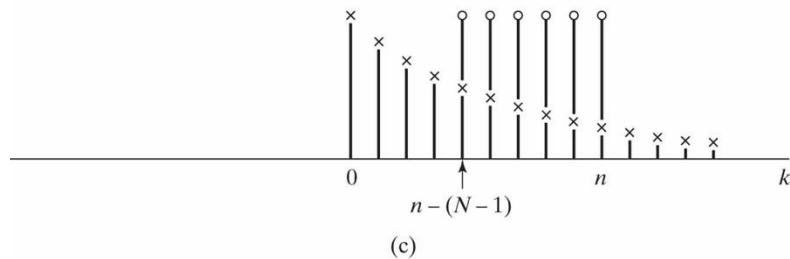
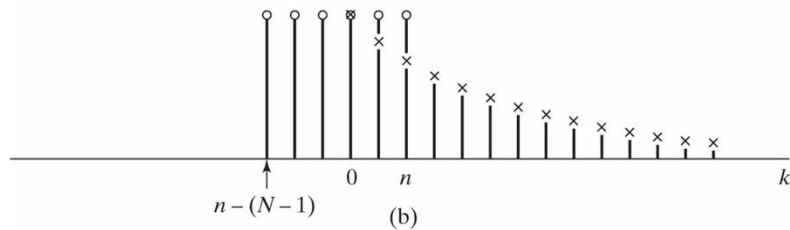
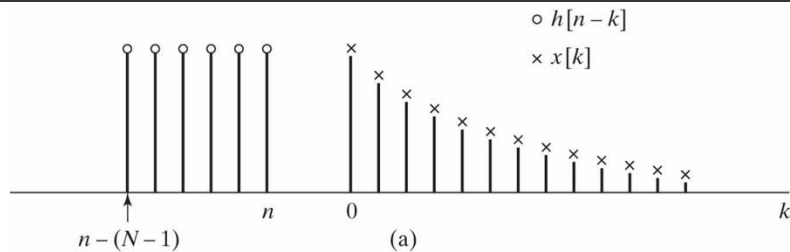
LINEAR TIME-INVARIANT SYSTEMS



$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Time-reverse, shift
origin by n

LINEAR TIME-INVARIANT SYSTEMS



Example:

$$x[n] = \alpha^n u[n]$$

$$h[n] = u[n] - u[n - N]$$

$$= \begin{cases} 1 & 0 \leq n \leq N - 1 \\ 0 & \text{else} \end{cases}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Analytical solution: [see notes](#)

LINEAR TIME-INVARIANT SYSTEMS

Properties of LTI systems follow from the properties of the convolution sum

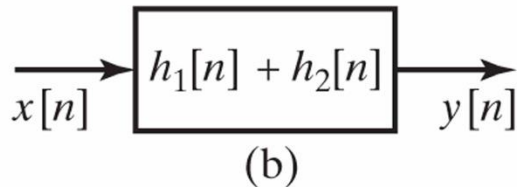
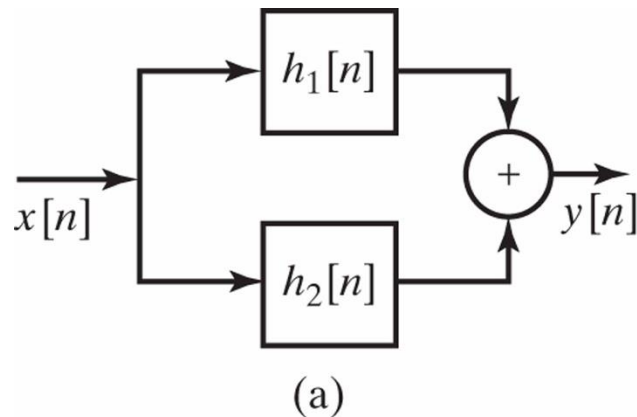
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$$

Property	Expressed in terms of convolution
Commutativity	$x[n] * h[n] = h[n] * x[n]$
Distributivity	$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$
Associativity	$\begin{aligned}(x[n] * h_1[n]) * h_2[n] &= x[n] * (h_1[n] * h_2[n]) \\ &= x[n] * (h_2[n] * h_1[n]) \\ &= (x[n] * h_2[n]) * h_1[n]\end{aligned}$

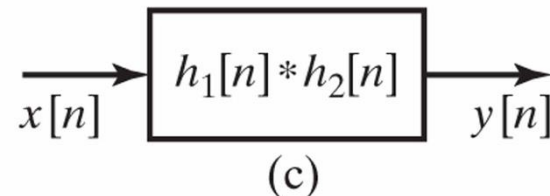
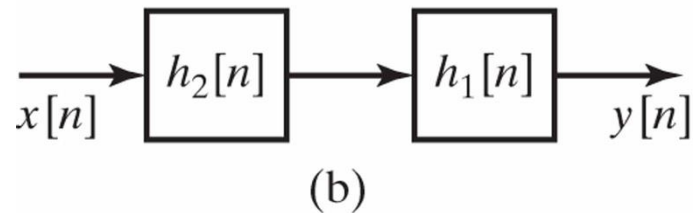
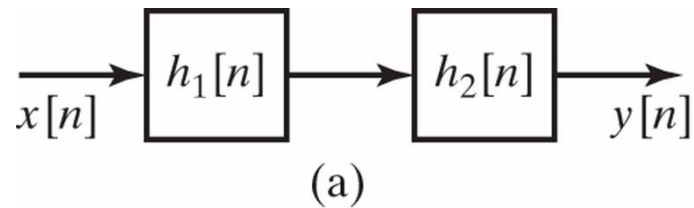
LINEAR TIME-INVARIANT SYSTEMS

Consequences of LTI system properties

Distributivity



Associativity



LINEAR TIME-INVARIANT SYSTEMS

- **Causality** in terms of impulse response: A LTI system is causal if and only if its impulse response is such that

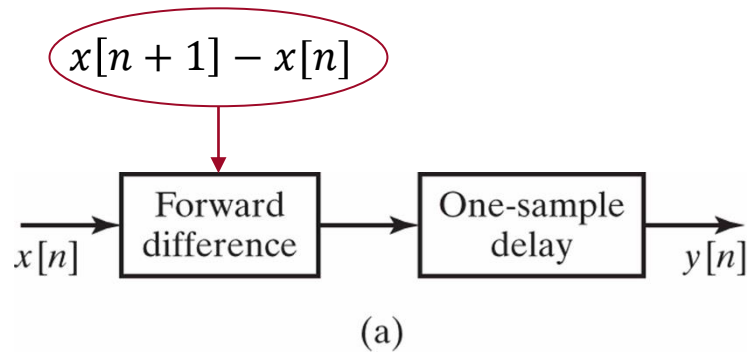
$$h[n] = 0, \quad \text{for } n < 0$$

- **Stability** in terms of impulse response: A LTI system is BIBO stable if and only if its impulse response is absolutely summable

$$B_h = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

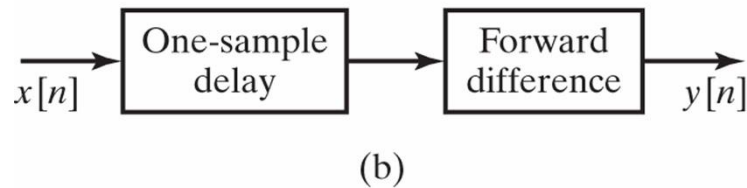
- See notes for proof

LINEAR TIME-INVARIANT SYSTEMS

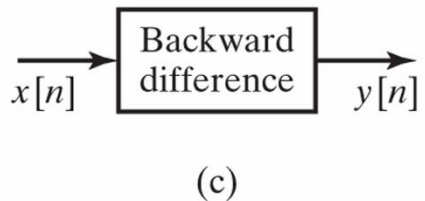


Delay by n_0 : $x[n] * \delta[n - n_0] = x[n - n_0]$

$$h[n] = (\delta[n + 1] - \delta[n]) * \delta[n - 1]$$

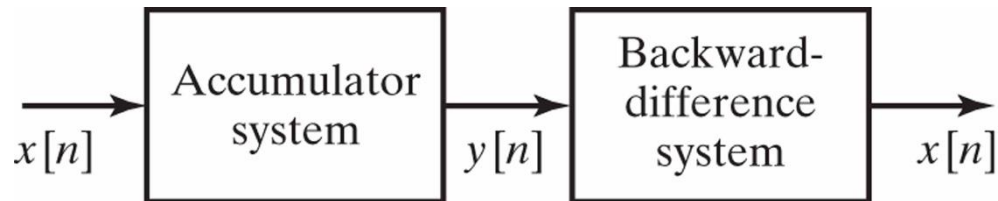


$$= \delta[n - 1] * (\delta[n + 1] - \delta[n])$$



$$= \delta[n] - \delta[n - 1]$$

LINEAR TIME-INVARIANT SYSTEMS



- Impulse response of the overall system:

$$\begin{aligned}h[n] &= u[n] * (\delta[n] - \delta[n - 1]) \\&= u[n] - u[n - 1] \\&= \delta[n]\end{aligned}$$

- So, the overall system is just an identity
- Hence, accumulator and backward-difference are **inverses** of each other

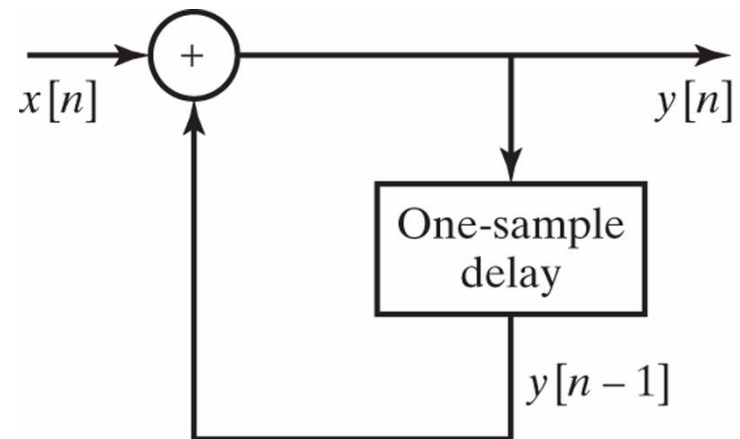
LINEAR TIME-INVARIANT SYSTEMS

- Many LTI systems can be represented as **difference equations**, which may suggest how they can be implemented
- For example, the accumulator system is defined as

$$y[n] = \sum_{k=-\infty}^n x[k]$$

- Rewriting:

$$\begin{aligned} y[n] &= x[n] + \sum_{k=-\infty}^{n-1} x[k] \\ &= x[n] + y[n-1] \end{aligned}$$



FREQUENCY-DOMAIN REPRESENTATION

- Recall from linear algebra: an **eigenvector** of a linear transformation (matrix) **A** is any vector **v** that satisfies

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

for some (possibly complex) λ , which is the corresponding **eigenvalue**

- Complex exponentials $e^{j\omega n}$ are **eigenfunctions** of LTI systems
- Let $x[n] = e^{j\omega n}$ be applied to a LTI system with impulse response $h[n]$, then

$$\begin{aligned} y[n] &= h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} \\ &= \sum_{k=-\infty}^{\infty} h[k]e^{j\omega n}e^{-j\omega k} = e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\ &= H(e^{j\omega})e^{j\omega n} \\ &= H(e^{j\omega})x[n] \end{aligned}$$

$H(e^{j\omega})$
(frequency response of the system)

DISCRETE-TIME FOURIER TRANSFORM (DTFT)

- Forward transform (time domain $\rightarrow$ frequency domain)

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

- Inverse transform (frequency domain $\rightarrow$ time domain)

$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega$$

- $H(e^{j\omega})$ is complex in general:

$$\begin{aligned} H(e^{j\omega}) &= H_R(e^{j\omega}) + j \cdot H_I(e^{j\omega}) \\ &= |H(e^{j\omega})| \angle H(e^{j\omega}) \end{aligned}$$

Real
part

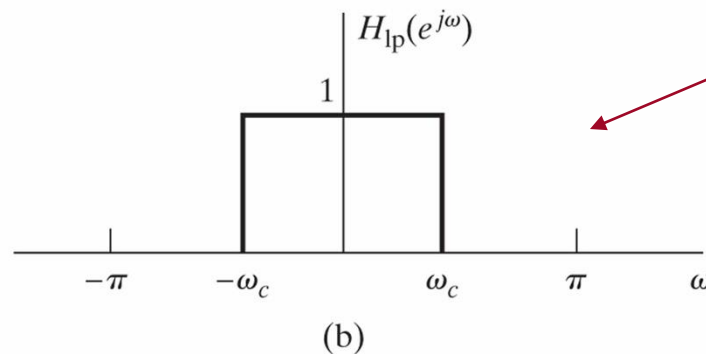
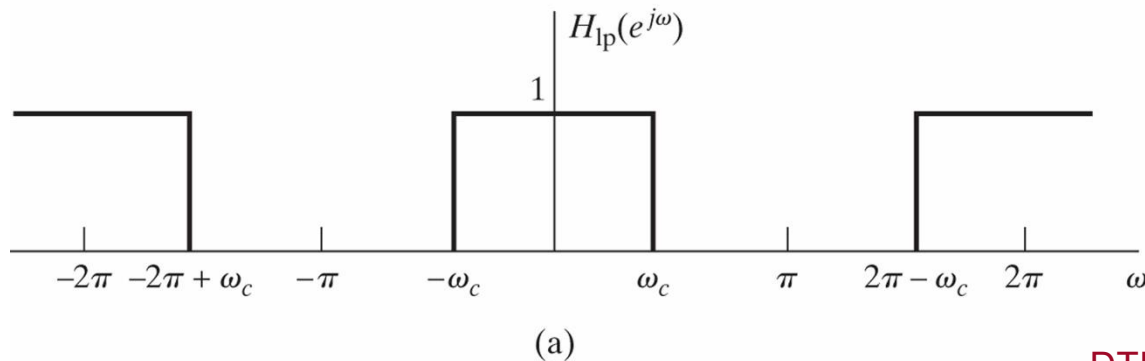
Imaginary
part

Magnitude

Phase

IDEAL FILTERS

- Filters are usually classified based on their magnitude response $|H(e^{j\omega})|$
- Example: ideal lowpass filter

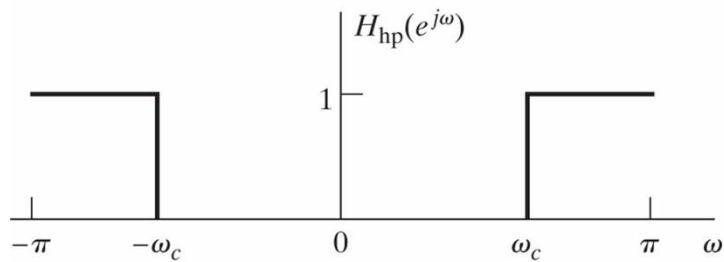


DTFT is 2π -periodic:

$$H(e^{j\omega}) = H(e^{j(\omega+2\pi k)})$$

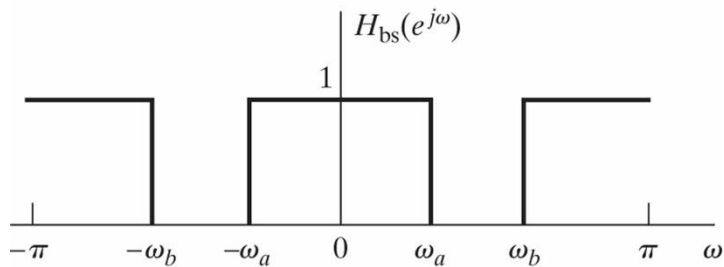
for any integer k . So it is sufficient to show $H(e^{j\omega})$ over an interval of length 2π , and this is usually the interval $[-\pi, \pi)$. Here, frequencies around 0 are **low frequencies**, and those around $\pm\pi$ are **high frequencies**.

IDEAL FILTERS



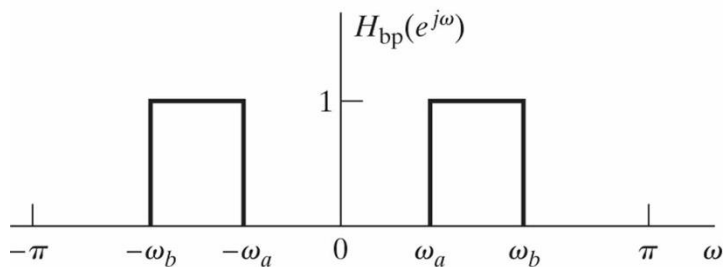
(a)

Ideal high-pass filter



(b)

Ideal band-stop filter



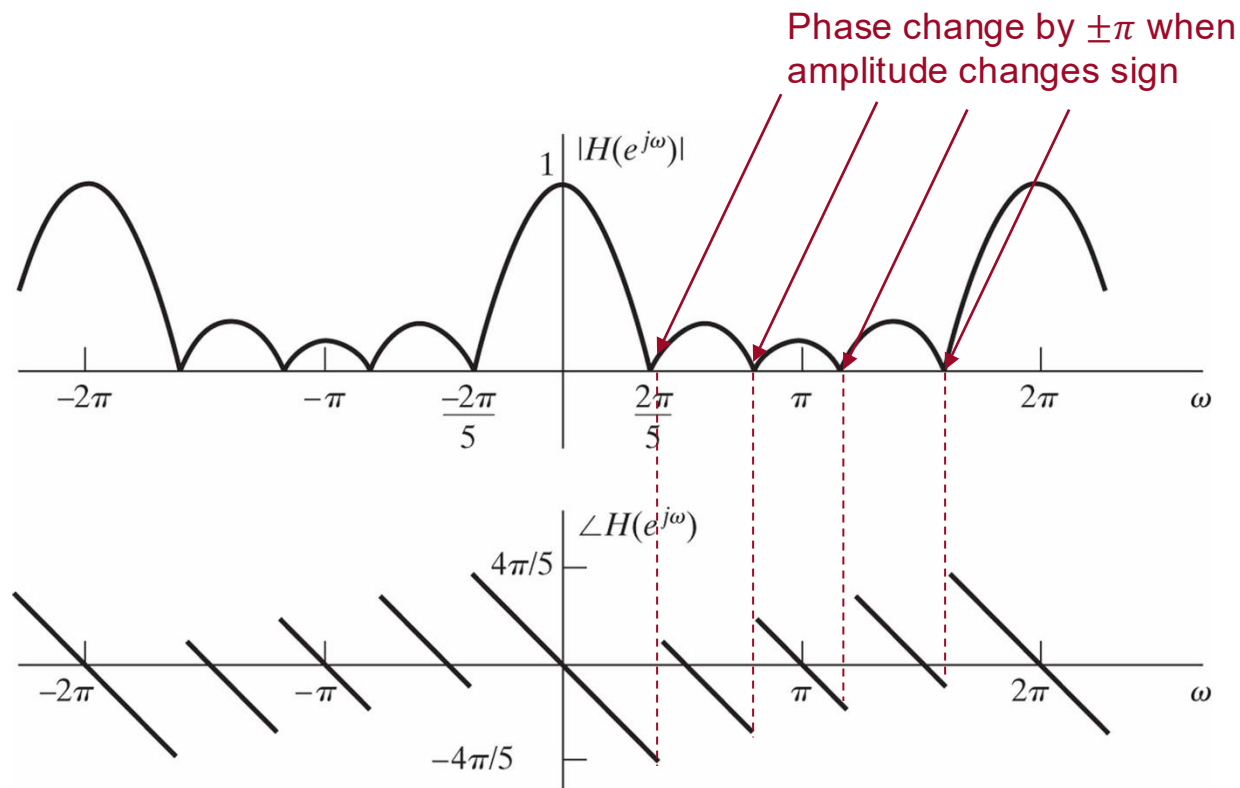
(c)

Ideal band-pass filter

FREQUENCY RESPONSE EXAMPLE

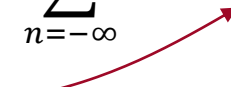
Example: frequency response of a moving average filter ([see notes](#) for details):

$$h[n] = \begin{cases} 1/5 & 0 \leq n \leq 4 \\ 0 & \text{else} \end{cases}$$



EXISTENCE OF DTFT

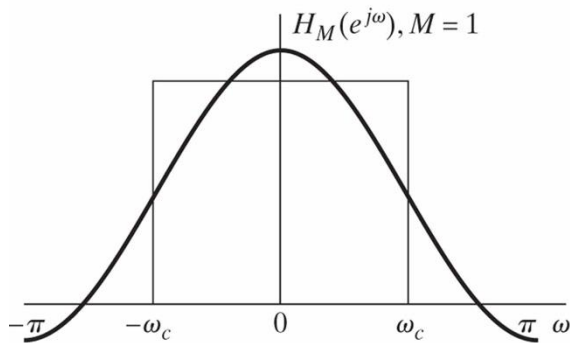
- When is DTFT well defined (i.e., $|X(e^{j\omega})| < \infty$) ?

$$\begin{aligned} |X(e^{j\omega})| &= \left| \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]e^{-j\omega n}| \\ &\leq \sum_{n=-\infty}^{\infty} |x[n]| \cdot |e^{-j\omega n}| = \sum_{n=-\infty}^{\infty} |x[n]| < \infty \end{aligned}$$


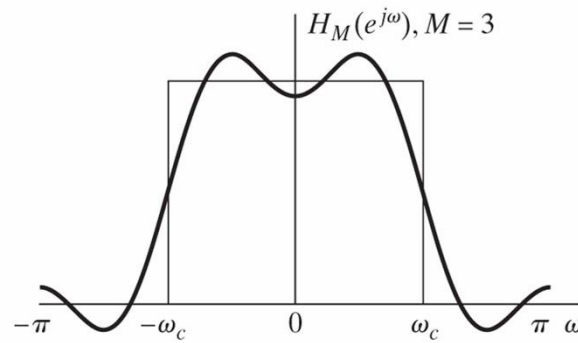
- So, when $x[n]$ is *absolutely summable*, its DTFT is guaranteed to exist
- This is a sufficient, but not a necessary condition
 - DTFT may exist even in cases where this condition is not met
 - Examples: ideal filters, constant sequences, complex exponential sequences

EXISTENCE OF DTFT

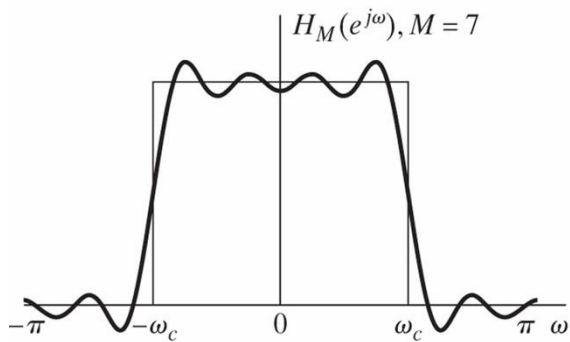
Example: ideal lowpass filter ([see notes](#) for details)



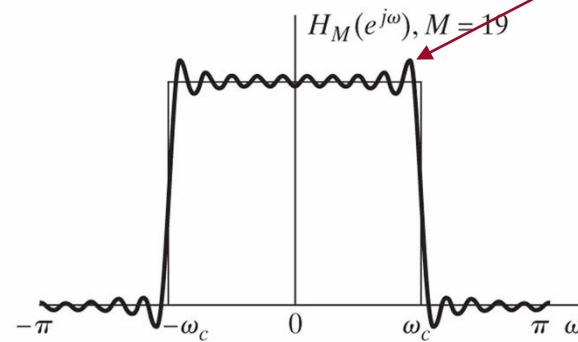
(a)



(b)



(c)



(d)

Gibbs phenomenon

EXISTENCE OF DTFT

Example: constant sequence ([see notes](#) for details)

$$x[n] = 1 \quad \Rightarrow \quad X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi \cdot \delta(\omega + 2\pi r)$$

Example: complex exponential sequence ([see notes](#) for details)

$$x[n] = e^{j\omega_0 n} \quad \Rightarrow \quad X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi \cdot \delta(\omega - \omega_0 + 2\pi r)$$

DTFT PROPERTIES

TABLE 2.1 SYMMETRY PROPERTIES OF THE FOURIER TRANSFORM

Sequence $x[n]$	Fourier Transform $X(e^{j\omega})$
1. $x^*[n]$	$X^*(e^{-j\omega})$
2. $x^*[-n]$	$X^*(e^{j\omega})$
3. $\mathcal{Re}\{x[n]\}$	$X_e(e^{j\omega})$ (conjugate-symmetric part of $X(e^{j\omega})$)
4. $j\mathcal{Im}\{x[n]\}$	$X_o(e^{j\omega})$ (conjugate-antisymmetric part of $X(e^{j\omega})$)
5. $x_e[n]$ (conjugate-symmetric part of $x[n]$)	$X_R(e^{j\omega}) = \mathcal{Re}\{X(e^{j\omega})\}$
6. $x_o[n]$ (conjugate-antisymmetric part of $x[n]$)	$jX_I(e^{j\omega}) = j\mathcal{Im}\{X(e^{j\omega})\}$
<i>The following properties apply only when $x[n]$ is real:</i>	
7. Any real $x[n]$	$X(e^{j\omega}) = X^*(e^{-j\omega})$ (Fourier transform is conjugate symmetric)
8. Any real $x[n]$	$X_R(e^{j\omega}) = X_R(e^{-j\omega})$ (real part is even)
9. Any real $x[n]$	$X_I(e^{j\omega}) = -X_I(e^{-j\omega})$ (imaginary part is odd)
10. Any real $x[n]$	$ X(e^{j\omega}) = X(e^{-j\omega}) $ (magnitude is even)
11. Any real $x[n]$	$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$ (phase is odd)
12. $x_e[n]$ (even part of $x[n]$)	$X_R(e^{j\omega})$
13. $x_o[n]$ (odd part of $x[n]$)	$jX_I(e^{j\omega})$

DTFT PROPERTIES

TABLE 2.2 FOURIER TRANSFORM THEOREMS

Sequence	Fourier Transform
$x[n]$	$X(e^{j\omega})$
$y[n]$	$Y(e^{j\omega})$
1. $ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
2. $x[n - n_d]$ (n_d an integer)	$e^{-j\omega n_d} X(e^{j\omega})$
3. $e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
4. $x[-n]$	$X(e^{-j\omega})$ $X^*(e^{j\omega})$ if $x[n]$ real.
5. $nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$
6. $x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
7. $x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Parseval's theorem:	
8. $\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) ^2 d\omega$	
9. $\sum_{n=-\infty}^{\infty} x[n]y^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})Y^*(e^{j\omega})d\omega$	

Linearity

Time shift

Frequency shift

Time reversal

Convolution in time domain

Convolution in frequency domain

DTFT PAIRS

TABLE 2.3 FOURIER TRANSFORM PAIRS

Sequence	Fourier Transform
1. $\delta[n]$	1
2. $\delta[n - n_0]$	$e^{-j\omega n_0}$
3. 1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
4. $a^n u[n]$ $(a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
5. $u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
6. $(n+1)a^n u[n]$ $(a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
7. $\frac{r^n \sin \omega_p(n+1)}{\sin \omega_p} u[n]$ $(r < 1)$	$\frac{1}{1 - 2r \cos \omega_p e^{-j\omega} + r^2 e^{-j2\omega}}$
8. $\frac{\sin \omega_c n}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1, & \omega < \omega_c, \\ 0, & \omega_c < \omega \leq \pi \end{cases}$
9. $x[n] = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
10. $e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$
11. $\cos(\omega_0 n + \phi)$	$\sum_{k=-\infty}^{\infty} [\pi e^{j\phi} \delta(\omega - \omega_0 + 2\pi k) + \pi e^{-j\phi} \delta(\omega + \omega_0 + 2\pi k)]$

READING ASSIGNMENT

- Read Sections 2.1– 2.9 in the textbook
- Start reading Chapter 3